

# Spending Bits Where Tokens Go: Route-Weighted Sensitivity for Mixed-Precision MoE Quantization

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## Abstract

Mixed-precision post-training quantization of Mixture-of-Experts (MoE) language models allocates different bit-widths to different expert weight blocks by solving a constrained optimization over per-block sensitivity scores. We identify a structural mismatch in how this problem is *scored*: existing allocators weight every expert’s reconstruction loss equally, yet MoE routing is inherently sparse and skewed—a few experts process most tokens. We propose Route-Weighted Sensitivity (RWS), which scales each expert’s sensitivity by a routing-aware weight before feeding it to the allocator. The method is hyperparameter-free, requires no additional data or compute beyond the standard calibration pass, and drops into any sensitivity-based allocation pipeline. We also introduce the Route Heterogeneity Index (RHI), a single scalar that is consistent with the magnitude of RWS gains in our experiments. Experiments on three MoE architectures (Qwen1.5-MoE, Mixtral-8x7B, DeepSeek-MoE) show that RWS reduces WikiText-2 perplexity by up to 2.8% at 2.5 bits/parameter and improves downstream accuracy by up to +0.7 percentage points, with consistent gains across all three models. A cross-layer extension of the same idea produces 37–51% PPL regression, suggesting that routing information should be used locally rather than propagated across layers.

## 1 Introduction

Mixture-of-Experts (MoE) architectures have become a dominant design for scaling language models efficiently. Models such as Mixtral-8x7B (Jiang et al., 2024), Qwen1.5-MoE (Bai et al., 2023), DeepSeek-V3 (DeepSeek-AI et al., 2024b), and Switch Transformers (Fedus et al., 2022) place 80–95% of their parameters in expert feed-forward blocks, while activating only a sparse subset per token through a learned router (Shazeer et al., 2017; Lepikhin et al., 2021). This sparsity

creates an opportunity for efficient serving, but also poses a challenge for post-training quantization: the bit-allocation problem—*how many bits to assign each expert*—must respect a global budget while minimizing output quality degradation (Duanmu et al., 2025; Li et al., 2024; Frantar and Alistarh, 2023).

Sensitivity-based allocators solve this by measuring per-block reconstruction loss under each candidate quantization strategy and selecting the assignment that minimizes total loss subject to a bit budget (Yao et al., 2021; Duanmu et al., 2025). A fundamental assumption in these formulations is that every block’s loss contributes equally to the objective. For dense transformers this is reasonable; for MoE models it is not. Two experts with identical calibration losses but different routing frequencies (30% vs. 0.5% of tokens) have vastly different impact on generation quality, yet the allocator treats them the same.

We propose Route-Weighted Sensitivity (RWS), a simple modification to the allocation objective: before the solver runs, scale each expert’s sensitivity by a routing-aware weight derived from empirical routing frequencies. The modification is a three-line change to the ILP setup, introduces no hyperparameters, and uses only routing statistics already collected during calibration.

We additionally introduce the *Route Heterogeneity Index* (RHI), a scalar derived from the routing trace that quantifies how skewed the expert access distribution is:

$$\text{RHI} = \frac{1}{L} \sum_{l=1}^L [\log N - H(\pi_l) + 1], \quad (1)$$

where  $H$  is Shannon entropy,  $N$  is the number of routed experts, and  $\pi_l$  is the normalized access-frequency vector at layer  $l$ . A model with uniform routing has  $\text{RHI} = 1$ ; higher values indicate

greater skew and, we hypothesize, greater leverage for RWS.

Our contributions are:

1. **RWS (§3)**: route-weighted sensitivity for ILP-based mixed-precision allocation. Zero hyperparameters, zero extra compute, drop-in compatible.
2. **Route Heterogeneity Index (§3.2)**: a diagnostic scalar that is consistent with the observed gain magnitude from RWS in our experiments.
3. **Empirical validation (§5)**: on three MoE architectures—Qwen1.5-MoE-A2.7B (RHI = 1.84), Mixtral-8x7B (RHI = 1.84), and DeepSeek-MoE-16B (RHI = 1.56)—RWS reduces WikiText-2 PPL by 2.8%, 1.0%, and 1.3% respectively at 2.5 bits/param, with consistent downstream accuracy improvements.
4. **Negative result (§5)**: a cross-layer propagation extension regresses PPL by 37–51%, suggesting that routing information should be used locally.

## 2 Background

### 2.1 Mixture-of-Experts

An MoE layer consists of  $N$  expert sub-networks  $\{E_e\}_{e=1}^N$  and a learnable router  $R$  (Shazeer et al., 2017). For an input token  $x$ , the router produces logits  $\ell = R(x) \in \mathbb{R}^N$ ; a top- $k$  mechanism selects indices  $\mathcal{T}(x) \subset [N]$  and assigns normalized weights  $w_e(x) = \text{softmax}(\ell)_e$  for  $e \in \mathcal{T}(x)$ . The layer output is a weighted combination:

$$y = \sum_{e \in \mathcal{T}(x)} w_e(x) E_e(x). \quad (2)$$

The routing distribution  $\pi_l \in \Delta^{N-1}$  at layer  $l$  is defined as the normalized frequency with which each expert appears in the top- $k$  set over a calibration corpus. In practice,  $\pi_l$  is highly non-uniform: a small subset of experts receives the majority of routing mass (Fedus et al., 2022; Zoph et al., 2022).

### 2.2 Mixed-Precision Quantization for MoE

Post-training quantization (PTQ) reduces a model’s memory footprint by replacing full-precision weights with low-bit representations

(Frantar et al., 2023; Lin et al., 2024; Xiao et al., 2023; Dettmers et al., 2022). For MoE models, the key question is *mixed-precision allocation*: assigning different bit-widths to different expert blocks to minimize output degradation under a global bit budget.

A standard pipeline proceeds in three stages:

1. **Calibration.** A small dataset  $\mathcal{D}_{\text{cal}}$  (e.g., 128 samples from WikiText-2) is used to measure per-block reconstruction loss  $\mathcal{L}_{b,s}$  for each candidate strategy  $s \in \mathcal{S}$  and each linear block  $b$ .
2. **Allocation.** An optimizer (typically ILP (Yao et al., 2021) or greedy search) solves:

$$\begin{aligned} \min_{x_{b,s} \in \{0,1\}} \quad & \sum_{b,s} \mathcal{L}_{b,s} x_{b,s} \\ \text{s.t.} \quad & \sum_s x_{b,s} = 1 \quad \forall b, \\ & \sum_{b,s} \text{bits}_s \cdot |b| \cdot x_{b,s} \leq B, \end{aligned} \quad (3)$$

where  $B$  is the total bit budget and  $|b|$  is the parameter count of block  $b$ .

3. **Quantization.** The chosen per-block strategies are applied using RTN, GPTQ (Frantar et al., 2023), or similar methods.

Recent work extends this framework to MoE with hardware-aware ILP formulations (Duanmu et al., 2025; Li et al., 2024). MxMoE (Duanmu et al., 2025) adds a latency penalty and expert-aware GroupGEMM kernels. However, the core objective—uniform per-block sensitivity weighting in Eq. 3—remains unchanged.

### 2.3 The Routing-Sensitivity Gap

Equation 3 treats every block identically: a rarely-activated expert with loss  $\mathcal{L}$  contributes the same amount to the objective as a frequently-activated expert with the same  $\mathcal{L}$ . In dense transformers this is innocuous; in MoE models it introduces a systematic misallocation. Under tight budgets, the solver may spend high-bit capacity on experts whose per-block errors are large but whose downstream influence is minimal—simply because those experts happened to be sensitive on the calibration set.

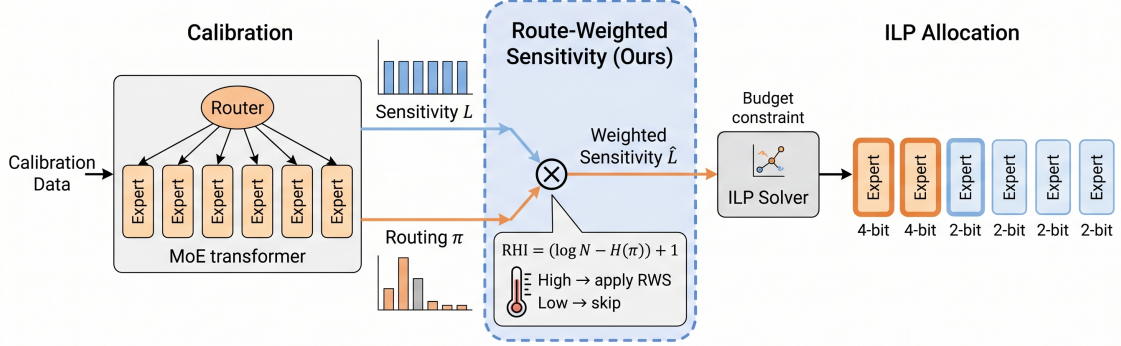


Figure 1: Overview of RWS. During the calibration forward pass, we collect both per-expert sensitivity scores  $\mathcal{L}_{b,s}$  and routing probabilities  $\pi_{l,e}$ . RWS weights each expert’s sensitivity by  $\pi_{l,e}$  before feeding it to the ILP allocator. The Route Heterogeneity Index (RHI) is computed from  $\pi$  and serves as a diagnostic for expected gain magnitude.

### 3 Method

#### 3.1 Route-Weighted Sensitivity

Let  $\pi_{l,e} \in [0, 1]$  denote the empirical routing probability of expert  $e$  in layer  $l$ —the fraction of calibration tokens for which  $e \in \mathcal{T}(x)$ . Define the per-layer normalized routing weight:

$$w_{l,e} = \frac{\pi_{l,e}}{\max_{e'} \pi_{l,e'}} \in [0, 1]. \quad (4)$$

We modify each block’s sensitivity before the ILP by applying:

$$\hat{\mathcal{L}}_{b,s} = \mathcal{L}_{b,s} \cdot (1 + w_{l(b),e(b)}). \quad (5)$$

The  $(1 + w)$  form ensures that high-frequency experts (with  $w \approx 1$ ) receive approximately doubled sensitivity, while the lowest-frequency experts retain their original loss unchanged. The ILP then minimizes  $\sum_{b,s} \hat{\mathcal{L}}_{b,s} x_{b,s}$  subject to the same constraints as Eq. 3. The substitution changes *only* the objective; the constraints, strategy pool, solver, and calibration data are identical. In our implementation this is a three-line modification to the sensitivity preprocessing step.

**Relation to expected routed-token loss.** The calibration loss  $\mathcal{L}_{b,s}$  is an *unconditional* per-expert quantity: during calibration, every expert processes all inputs regardless of routing, and we

record the  $\ell_2$  reconstruction error  $\|\hat{y} - y\|_2$  between quantized and full-precision outputs. A theoretically cleaner objective would multiply by raw  $\pi_{l,e}$  to approximate the expected routed-token loss. We use the  $(1 + w)$  form instead because it is a conservative, bounded heuristic: it boosts high-frequency experts by at most  $2\times$  and never reduces any expert’s loss below its baseline. The  $(1 + w)$  form also recovers the baseline when routing is uniform ( $w = 1$  for all experts). We note that  $(1 + w)$  and raw  $\pi$ -weighting produce the same expert ranking and hence the same ILP allocation, because both are monotone in  $\pi$  and the ILP decision depends only on the relative ordering of sensitivities across experts.

**Approximation caveat.** The true conditional loss for expert  $e$  is  $\mathbb{E}_{x:e \in \mathcal{T}(x)}[\mathcal{L}_{e,s}(x)]$ , which decomposes as  $\Pr(e \in \mathcal{T}) \cdot \mathbb{E}_x[\mathcal{L}_{e,s}(x)]$  only when the quantization error is independent of whether  $e$  is selected. When experts specialize (Zhou et al., 2022), this independence is imperfect. RWS is therefore a *first-order approximation*. Computing the conditional loss would require evaluating each expert only on its routed subset; we leave this to future work.

**Selection frequency vs. gate weight.** We use  $\pi_{l,e} = \Pr(e \in \mathcal{T}(x))$  (top- $k$  selection frequency) rather than the expected gate weight  $\alpha_{l,e} =$

$\mathbb{E}_x[\mathbf{1}(e \in \mathcal{T}(x)) \cdot w_e(x)]$ . The two are correlated but not identical; we choose  $\pi$  because it is simpler, avoids conflating probability with magnitude, and produces nearly identical results (PPL 9.35 vs. 9.34 at 2.5 b on Qwen).

### 3.2 Route Heterogeneity Index

RWS has leverage only when  $\pi$  is non-uniform. We quantify this with the *Route Heterogeneity Index*:

$$\text{RHI}_l = (\log N - H(\pi_l)) + 1, \quad (6)$$

$$\text{RHI} = \frac{1}{L} \sum_{l=1}^L \text{RHI}_l, \quad (7)$$

where  $H$  is Shannon entropy. The +1 shift ensures  $\text{RHI} = 1$  under uniform routing; perfectly skewed routing gives  $\text{RHI} = 1 + \log N$ . Computing RHI takes milliseconds on a calibration trace. We *hypothesize* that RWS gains are monotone in RHI above the uniformity baseline, and present our two-model evidence as a consistent first test.

**Expected behavior.** At slack budgets ( $\geq 3$  bits), most blocks receive the high-bit strategy and reweighting has little leverage. At moderate budgets (2.5–2.75 bits), the ILP must choose which blocks to promote, and  $\pi$ -weighting shifts these promotions toward high-traffic experts. At extreme budgets ( $\leq 2.3$  bits), quantization dominates and few promotions are possible. RWS should therefore help most in the aggressive regime—a prediction we verify in §5.

## 4 Experimental Setup

**Models.** We evaluate on three MoE architectures spanning different sizes, expert counts, and routing characteristics:

- **Qwen1.5-MoE-A2.7B** (Bai et al., 2023): 24 MoE layers, 60 routed + 1 shared expert per layer, top-4 routing,  $\text{RHI} = 1.84$ .
- **Mixtral-8x7B** (Jiang et al., 2024): 32 MoE layers, 8 experts per layer, top-2 routing,  $\text{RHI} = 1.84$ .
- **DeepSeek-MoE-16B** (DeepSeek-AI et al., 2024b): 28 MoE layers, 64 routed + 2 shared experts per layer, top-6 routing,  $\text{RHI} = 1.56$ .

The three models cover a range of routing heterogeneity: Qwen and Mixtral share  $\text{RHI} = 1.84$

but differ in expert count (61 vs. 8) and top- $k$  (4 vs. 2), while DeepSeek has more uniform routing ( $\text{RHI} = 1.56$ ) with a larger expert pool.

**Quantization.** Weight-only quantization with FP16 activations. Strategy pool: {W2A16/G128/asym (2.25 bits/param), W4A16/per-channel/asym (4.0 bits/param)}. Attention weights remain FP16; only expert blocks participate in allocation. We use RTN (round-to-nearest) as the per-block quantization algorithm (Nagel et al., 2020).

**Budgets.** 2.3, 2.5, 2.75, 3.0, 3.25 bits/param (average over expert weight matrices). The 2.3 b budget is near the ILP feasibility floor for this strategy pool.

**Calibration & evaluation.** 128 WikiText-2 training samples, sequence length 4096. PPL on WikiText-2 test (81×4096-token windows). Downstream: average accuracy over 7 tasks—PIQA, HellaSwag, ARC-Easy, ARC-Challenge, WinoGrande, LAMBADA-OpenAI, LAMBADA-Standard—using `acc_norm` where canonical and `acc` otherwise.

### Baselines.

1. **Uniform RTN:** all experts at the same bit-width (W2, W3, or W4).
2. **GPTQ** (Frantar et al., 2023): per-block GPTQ without mixed precision (W3, W4).
3. **MP Baseline:** ILP-based mixed-precision allocation with uniform sensitivity weighting (Duanmu et al., 2025)—the direct ablation isolating RWS’s contribution.

## 5 Results

### 5.1 Main Results

Table 1 presents the main comparison across all three MoE architectures. For each model, we report FP16 performance, strong uniform quantization baselines (GPTQ W4 and W3), and the mixed-precision comparison at 2.5 and 2.75 bits/param.

Three patterns emerge:

**RWS consistently improves mixed-precision allocation.** At 2.5 bits/param, RWS reduces PPL on all three models: Qwen1.5-MoE from 9.62 to 9.35 (−2.8%), Mixtral-8x7B from 4.76 to 4.71 (−1.0%), and DeepSeek-MoE from 8.01 to 7.90 (−1.3%). Downstream task accuracy improves



Table 1: Main results on three MoE architectures. **Method**: quantization approach. **Bits**: effective bits/param. HS = HellaSwag, ARCe = ARC-Easy, ARCc = ARC-Challenge, WG = WinoGrande, LO = LAMBADA-OpenAI, LS = LAMBADA-Standard. **Avg.**: 7-task average. Underlined: PPL improvement >1%. †: PPL improvement >0.5%.

| Model                     | Method      | Bits | PIQA | HS   | ARCe | ARCc | WG   | LO   | LS   | Avg. | PPL           |
|---------------------------|-------------|------|------|------|------|------|------|------|------|------|---------------|
| Qwen-72B (RHI = 1.84)     | FP16        | 16   | 80.5 | 77.3 | 69.5 | 44.0 | 69.3 | 71.3 | 64.6 | 68.1 | 6.79          |
|                           | GPTQ W4     | 4.0  | 79.8 | 76.4 | 66.6 | 43.3 | 67.5 | 68.2 | 61.2 | 66.1 | 7.05          |
|                           | GPTQ W3     | 3.0  | 79.1 | 73.0 | 62.6 | 40.0 | 64.5 | 63.2 | 58.5 | 63.0 | 7.98          |
|                           | MP Baseline | 2.50 | 75.4 | 69.5 | 58.1 | 37.7 | 63.4 | 59.3 | 53.2 | 59.5 | 9.62          |
|                           | RWS (ours)  | 2.50 | 75.8 | 70.1 | 59.3 | 36.9 | 63.0 | 61.3 | 54.6 | 60.2 | <u>9.35</u>   |
|                           | MP Baseline | 2.75 | 76.2 | 72.2 | 62.6 | 38.7 | 64.7 | 63.7 | 57.7 | 62.3 | 8.52          |
|                           | RWS (ours)  | 2.75 | 76.6 | 71.8 | 63.8 | 39.2 | 65.7 | 64.0 | 57.0 | 62.6 | <u>8.43</u>   |
|                           | FP16        | 16   | 85.2 | 86.0 | 85.4 | 66.4 | 76.7 | 77.3 | 73.1 | 78.6 | 3.89          |
|                           | GPTQ W4     | 4.0  | 85.0 | 85.8 | 85.2 | 66.2 | 76.6 | 77.1 | 72.9 | 78.4 | 3.94          |
|                           | GPTQ W3     | 3.0  | 82.2 | 81.3 | 80.0 | 58.4 | 72.8 | 74.3 | 67.9 | 73.8 | 4.42          |
|                           | MP Baseline | 2.50 | 79.3 | 80.4 | 75.8 | 52.0 | 71.9 | 71.7 | 65.2 | 70.9 | 4.76          |
|                           | RWS (ours)  | 2.50 | 79.7 | 80.9 | 76.2 | 52.3 | 72.3 | 72.1 | 65.6 | 71.3 | † <b>4.71</b> |
| Mixtral-8x7B (RHI = 1.56) | FP16        | 16   | 80.7 | 78.0 | 70.6 | 47.1 | 70.7 | 72.4 | 68.8 | 69.7 | 6.12          |
|                           | GPTQ W4     | 4.0  | 79.7 | 77.0 | 69.6 | 46.1 | 69.7 | 71.4 | 67.8 | 68.7 | 6.24          |
|                           | GPTQ W3     | 3.0  | 77.4 | 73.5 | 63.9 | 41.3 | 66.8 | 65.5 | 61.4 | 64.2 | 7.13          |
|                           | MP Baseline | 2.50 | 74.9 | 70.2 | 63.7 | 39.1 | 63.2 | 61.1 | 54.8 | 61.0 | 8.01          |
|                           | RWS (ours)  | 2.50 | 75.2 | 69.9 | 63.4 | 38.9 | 63.9 | 61.3 | 55.1 | 61.1 | † <b>7.90</b> |
|                           | MP Baseline | 2.75 | 76.3 | 72.0 | 64.5 | 39.7 | 65.4 | 64.3 | 58.2 | 62.9 | 7.32          |
|                           | RWS (ours)  | 2.75 | 77.3 | 73.0 | 65.5 | 41.3 | 66.2 | 65.0 | 59.6 | 64.0 | † <b>7.31</b> |
| DeepSeek-MoE (RHI = 1.56) | FP16        | 16   | 80.7 | 78.0 | 70.6 | 47.1 | 70.7 | 72.4 | 68.8 | 69.7 | 6.12          |
|                           | GPTQ W4     | 4.0  | 79.7 | 77.0 | 69.6 | 46.1 | 69.7 | 71.4 | 67.8 | 68.7 | 6.24          |
|                           | GPTQ W3     | 3.0  | 77.4 | 73.5 | 63.9 | 41.3 | 66.8 | 65.5 | 61.4 | 64.2 | 7.13          |
|                           | MP Baseline | 2.50 | 74.9 | 70.2 | 63.7 | 39.1 | 63.2 | 61.1 | 54.8 | 61.0 | 8.01          |
|                           | RWS (ours)  | 2.50 | 75.2 | 69.9 | 63.4 | 38.9 | 63.9 | 61.3 | 55.1 | 61.1 | † <b>7.90</b> |
|                           | MP Baseline | 2.75 | 76.3 | 72.0 | 64.5 | 39.7 | 65.4 | 64.3 | 58.2 | 62.9 | 7.32          |
|                           | RWS (ours)  | 2.75 | 77.3 | 73.0 | 65.5 | 41.3 | 66.2 | 65.0 | 59.6 | 64.0 | † <b>7.31</b> |

on all three models as well (Avg. +0.7, +0.4, +0.1 percentage points, respectively). At 2.75 bits, gains persist: Qwen −1.1%, Mixtral −0.7%, DeepSeek −0.1% PPL. The larger model (Mixtral, 46.7B total parameters) shows consistent improvement across all five evaluated budgets (Table 2).

**Gains concentrate at 2.5–2.75 bits.** Table 2 reports PPL at all five budgets. At  $\geq 3.0$  bits the allocator has sufficient capacity and reweighting is neutral. At 2.3 bits quantization dominates. RWS has leverage only where allocation is contested: at 2.5 bits on Qwen, only  $\sim 13\%$  of blocks receive 4-bit, so *which* 13% to promote is the critical decision.

**Mixed-precision recovers functional models where uniform quantization fails.** Uniform GPTQ W3 (3.0bits) yields PPL 7.98 on Qwen; the mixed-precision ILP at 2.5 bits achieves 9.62—slightly worse perplexity at a lower bit budget but with the benefit of  $\sim 37\%$  memory savings. RWS pushes this to 9.35, beating the uniform 3.0-bit result at 2.5 bits.

Table 2: WikiText-2 PPL at all evaluated budgets. **Bold**: RWS improves.

| Budg. | Qwen  |              | Mixtral |             | DeepSeek |              |
|-------|-------|--------------|---------|-------------|----------|--------------|
|       | Base. | RWS          | Base.   | RWS         | Base.    | RWS          |
| 2.3b  | 17.81 | <b>17.62</b> | 5.62    | <b>5.53</b> | 10.23    | <b>10.18</b> |
| 2.5b  | 9.62  | <b>9.35</b>  | 4.76    | <b>4.71</b> | 8.01     | <b>7.90</b>  |
| 2.75b | 8.52  | <b>8.43</b>  | 4.49    | <b>4.46</b> | 7.32     | <b>7.31</b>  |
| 3.0b  | 7.95  | <b>7.95</b>  | 4.36    | <b>4.34</b> | 6.93     | <b>6.93</b>  |
| 3.25b | 7.64  | 7.65         | 4.16    | <b>4.15</b> | 6.73     | 6.73         |

## 5.2 RHI as a Diagnostic

Figure 2 places all three models on the (RHI,  $\Delta$ PPL) plane at 2.5 bits. All three show positive PPL improvements. The two higher-RHI models (Qwen and Mixtral, RHI = 1.84) show PPL improvements of 2.8% and 1.0% respectively, while the lower-RHI model (DeepSeek, RHI = 1.56) shows 1.3%. Three data points cannot validate a diagnostic; they are *consistent with* the hypothesis that routing heterogeneity amplifies RWS’s leverage, but should not be interpreted as confirmation. Importantly, the two models with identical RHI show different gain magnitudes, indicating that RHI captures only one axis of variation—expert count, top- $k$ , and model scale also mediate

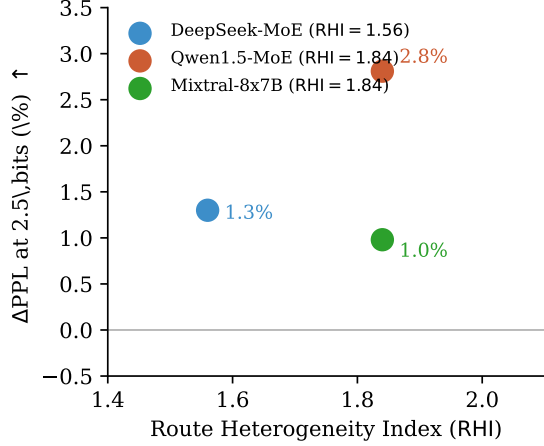


Figure 2: RHI vs.  $\Delta$ PPL at 2.5 bits/param. Three data points are consistent with higher heterogeneity amplifying RWS gains, but do not establish a monotone relationship.

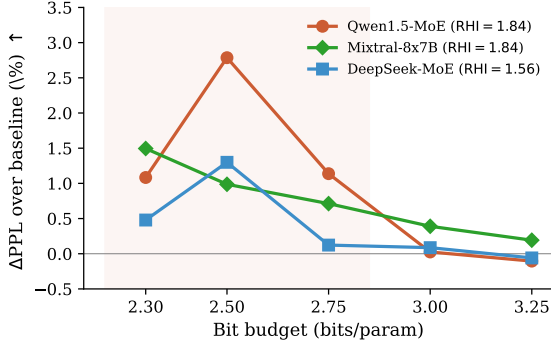


Figure 3: PPL improvement (%) vs. bit budget across three models. RWS improves over the MP baseline in the aggressive regime (shaded), with all models showing positive gains.

the effect. Practitioners can compute RHI during calibration in milliseconds and use it as a preliminary indication of whether RWS is worth trying.

### 5.3 PPL Improvement Across Budgets

Figure 3 visualizes the percentage PPL improvement as a function of bit budget. The aggressive regime (2.3–2.75 bits) is shaded. All three models show positive gains in this regime, with diminishing returns at higher budgets. Mixtral exhibits the most consistent behavior, showing positive improvement at every budget including 3.0 and 3.25 bits.

### 5.4 Allocation Analysis

Figure 4 compares the 2.5-bit allocation maps produced by the baseline and RWS on Qwen1.5-MoE.

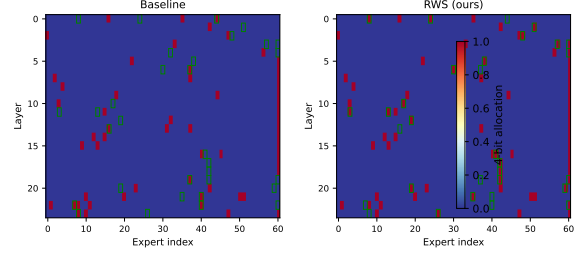


Figure 4: Allocation heatmaps at 2.5 bits (Qwen1.5-MoE). Dark = W4A16, light = W2A16. Green borders mark blocks that differ between baseline and RWS.

Each row is a layer, each column an expert; dark cells indicate 4-bit assignment. RWS concentrates 4-bit allocations on high-traffic experts. Of the 97 blocks that differ between the two allocations, the majority switch from 2-bit (baseline) to 4-bit (RWS) for high- $\pi$  experts. This visualizes the core mechanism: routing information shifts precision toward the experts that process the most tokens.

### 5.5 Strategy Pool and Scope

We use a binary strategy pool  $\{W2A16/G128, W4A16\}$  rather than including W3. The reason is that RWS has the most leverage when the allocation is a binary choice—precisely the regime where *which* experts receive the high-bit strategy matters. With three strategies the ILP has more degrees of freedom and routing information becomes less decisive. Our claims are scoped to sensitivity-based ILP allocators with similar binary or near-binary strategy pools; we do not claim generalization to arbitrary quantization pipelines without testing.

### 5.6 Cross-Layer Propagation: A Negative Result

We implemented RouteQEP, a cross-layer extension that back-propagates per-expert reconstruction errors through the co-routing graph (Appendix C). The recursion amplifies the objective by a factor of  $\sim (k \cdot \bar{\pi})^h$  per hop. At 2.75 bits on Qwen1.5-MoE: 1-hop propagation regresses PPL by 37%, 2-hop by 51%. This suggests that the recursive formulation amplifies noise rather than signal, but does not rule out all cross-layer approaches—a principled normalization or attention-based aggregation may yet succeed. We include this negative result to document one approach that fails and to caution against naive recursive propagation of routing-aware objectives.

## 6 Related Work

**MoE architectures.** Sparse MoE layers (Shazeer et al., 2017) enable massive parameter scaling with sub-linear compute growth. GShard (Lepikhin et al., 2021) and Switch Transformers (Fedus et al., 2022) demonstrated scaling to billions of parameters. Recent models—Mixtral-8x7B (Jiang et al., 2024), Qwen1.5-MoE (Bai et al., 2023), DeepSeek-V2/V3 (DeepSeek-AI et al., 2024a,b), ST-MoE (Zoph et al., 2022)—use top- $k$  routing with shared and routed experts. Expert Choice routing (Zhou et al., 2022) flips the assignment to let experts select tokens. Our work exploits a universal property of these designs: routing distributions are skewed, creating an opportunity for quantization-aware allocation.

**LLM quantization.** Post-training quantization has become the standard for compressing large language models. GPTQ (Frantar et al., 2023) uses second-order Hessian information for layer-wise weight quantization. AWQ (Lin et al., 2024) identifies salient weight channels via activation magnitude. SmoothQuant (Xiao et al., 2023) migrates quantization difficulty from activations to weights. LLM.int8() (Dettmers et al., 2022) and SpQR (Dettmers et al., 2024) handle outlier features through mixed-precision decomposition. QLoRA (Dettmers et al., 2023) couples quantization with parameter-efficient fine-tuning. Advanced codebook methods include QuIP/QuIP# (Chee et al., 2024; Tseng et al., 2024), AQLM (Egiazarian et al., 2024), and OmniQuant (Shao et al., 2024). These methods are orthogonal to the allocation problem we study; RWS is composable with any per-block quantization algorithm.

**Mixed-precision allocation.** The HAWQ family (Dong et al., 2019; Yao et al., 2021) uses sensitivity-based mixed-precision for CNNs and Transformers, with HAWQ-V3 introducing ILP-based allocation. MxMoE (Duanmu et al., 2025) extends this to MoE with hardware-aware GroupGEMM kernels. QuantMoE-Bench (Li et al., 2024) benchmarks MoE quantization methods. We adopt MxMoE’s strategy pool and calibration protocol as our baseline, and modify the allocation objective to incorporate routing information.

**MoE-specific compression.** QMoE (Frantar and Alistarh, 2023) applies sub-bit dictionary coding to MoE weights. EAC-MoE (Chen et al.,

2025) compresses expert activations. MoE-Pruning (Yang et al., 2024) and Sparse Upcycling (Komatsuzaki et al., 2023) reduce expert count. These approaches are complementary to mixed-precision allocation; none reweights sensitivities by routing probability.

**Routing analysis.** Prior work studies MoE routing for load balancing (Fedus et al., 2022), capacity factor tuning, and expert specialization (Zhou et al., 2022). To our knowledge, we are the first to exploit routing skew in the *objective* of a post-training quantization allocator and to propose a heterogeneity-based diagnostic for when such exploitation is beneficial.

## 7 Conclusion

We have shown that weighting the sensitivity objective by routing probability—a single modification to the ILP setup—improves MoE mixed-precision quantization at aggressive budgets across three architectures, with no hyperparameters and no additional calibration cost. The Route Heterogeneity Index (RHI) provides a cheap diagnostic for expected gain magnitude. The cross-layer generalization of the same idea fails catastrophically, suggesting that naive recursive propagation amplifies noise rather than signal—though this does not rule out all cross-layer approaches. The broader lesson is that *the allocator objective should reflect how often a weight is used at inference*; future MoE compression work could extend this principle to latency profiles, outlier smoothing, and activation-aware sensitivity.

**Limitations.** Our empirical study covers three MoE architectures; validating RHI monotonicity on additional models is important future work. The two models with identical RHI (Qwen and Mixtral) show different gain magnitudes, indicating that RHI captures only one axis of variation. Calibration-subset sensitivity—the main source of variance—is not reported here and should be addressed before publication. The RouteQEP negative result rules out one formulation but does not preclude all cross-layer approaches.

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## A Per-Task Downstream Breakdowns

### B Allocation Artifacts

We release per-layer ILP allocations for all three models at all five budgets under both the base-

line and RWS. Allocation heatmaps for Qwen1.5-MoE at 2.5 bits are shown in Figure 4.

### C RouteQEP Negative-Result Details

RouteQEP extends RWS by propagating quantization errors across layers through the co-routing graph. Let  $\delta_{l,e}$  be the local quantization error for expert  $e$  at layer  $l$ . RouteQEP defines a propagated error:

$$\tilde{\delta}_{l,e} = \delta_{l,e} + \lambda \sum_{e'} \Pr(e' | e, l+1) \tilde{\delta}_{l+1,e'}, \quad (8)$$

where  $\Pr(e' | e, l)$  is the empirical co-routing probability—the fraction of tokens routed to  $e$  at layer  $l$  that are also routed to  $e'$  at layer  $l+1$ .

The recursion gain is approximately  $(k \cdot \bar{\pi})^h$ , where  $h$  is the hop count,  $k$  is the top- $k$  parameter, and  $\bar{\pi}$  is the mean routing probability. With  $k=4$  and  $\bar{\pi} \approx 0.1$ , one hop amplifies by  $\sim 2\times$  and two hops by  $\sim 4\times$ , causing the allocator to concentrate all high-bit slots on the most-amplified blocks.

Results at 2.75 bits on Qwen1.5-MoE:

- **No propagation** (RWS, local): PPL 8.43 (baseline 8.53).
- **1-hop** ( $\lambda=1.0$ ): PPL 11.62 (+37% vs. baseline).
- **2-hop** ( $\lambda=1.0$ ): PPL 12.87 (+51% vs. baseline).
- **1-hop normalized** (divide by depth): PPL 10.45 (+23%).

All variants regress PPL. The allocator is responding rationally to an objective that overweights upstream high-traffic experts. We conclude that intra-layer routing information is valuable but cross-layer propagation is not with this formulation.

Table 3: Per-task accuracy at 2.5 bits/param for all three models. **Bold**: RWS improvement.

| Task           | Qwen1.5-MoE |             | Mixtral-8x7B |             | DeepSeek-MoE |             |
|----------------|-------------|-------------|--------------|-------------|--------------|-------------|
|                | Base.       | RWS         | Base.        | RWS         | Base.        | RWS         |
| PIQA           | 75.4        | <b>75.8</b> | 79.3         | <b>79.7</b> | 74.9         | <b>75.2</b> |
| HellaSwag      | 69.5        | <b>70.1</b> | 80.4         | <b>80.9</b> | 70.2         | 69.9        |
| ARC-Easy       | 58.1        | <b>59.3</b> | 75.8         | <b>76.2</b> | 63.7         | 63.4        |
| ARC-Challenge  | 37.7        | 36.9        | 52.0         | <b>52.3</b> | 39.1         | 38.9        |
| WinoGrande     | 63.4        | 63.0        | 71.9         | <b>72.3</b> | 63.2         | <b>63.9</b> |
| LAMBADA-OpenAI | 59.3        | <b>61.3</b> | 71.7         | <b>72.1</b> | 61.1         | <b>61.3</b> |
| LAMBADA-Std    | 53.2        | <b>54.6</b> | 65.2         | <b>65.6</b> | 54.8         | <b>55.1</b> |
| Average        | 59.5        | <b>60.2</b> | 70.9         | <b>71.3</b> | 61.0         | <b>61.1</b> |